

Total No. of Questions : 10]

SEAT No. :

P1854

[4859]-1015

[Total No. of Pages : 3

B.E. (Mechanical)

**CAD / CAM AND AUTOMATION
(2012 Course) (Semester-I) (End Sem)**

Time : 2.30 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answers in one answer book.*
- 2) *Answer Q. 1 or Q. 2, Q. 3 or Q. 4, Q. 5 or Q. 6, Q. 7 or Q. 8, Q. 9 or Q. 10.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Figures to the right side indicate full marks.*
- 5) *Use of calculator is allowed.*
- 6) *Assume suitable data if necessary.*

Q1) a) Line A(5, 5) B(10, 15) is to be rotated about point B by 60° in CCW direct find the new position of point A and B of line. [6]

b) In concatenated transformation why translation matrix to be written in homogeneous form also write translation matrix in homogeneous form. [4]

OR

Q2) a) Explain Geometric translational and rotational mapping and its need. [6]

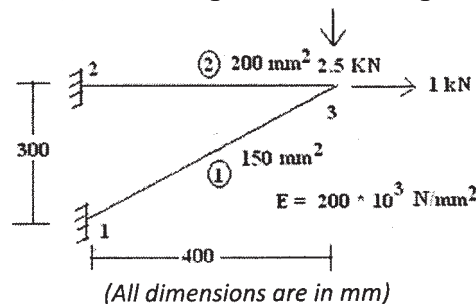
b) Explain Bezier synthetic curve. [4]

Q3) a) Explain CSG technique of solid modeling and its advantages and limitation. [6]

b) Explain the concept of shape function for 1-D element. [4]

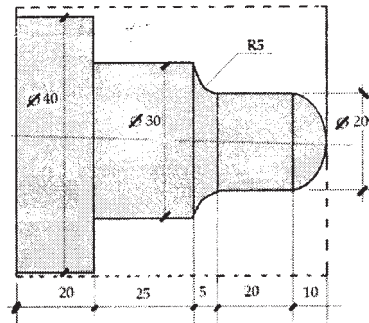
OR

Q4) For the truss element shown in the figure find Global stiffness matrix and write in the form $KQ = F$. Compute nodal displacements. [10]



P.T.O.

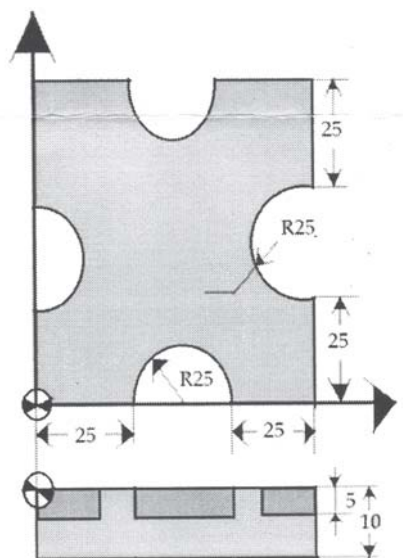
- Q5) a)** Explain the canned cycle for Drilling, Boring and Tapping. [8]
- b)** Write CNC part program for roughing and finishing using canned cycle for turned component as shown in figure. Assume Suitable cutting data. [10]



(All dimensions are in mm)

OR

- Q6) a)** Explain the concept of tool radius compensation while CNC programming and explain G codes and its format for cutter compensation. [6]
- b)** Write CNC part program for Milling and Drilling for component as shown in figure use concept of sub programming so that depth of cut per pass is 1 mm. Assume Suitable cutting data. [12]



(All dimensions are in mm)

- Q7) a)** Explain Rapid Prototyping (RP) systems in detail. List various RP techniques. Explain 3-D Printing with neat sketch. [8]
- b)** Explain Rapid Tooling in detail. List different Rapid Tooling methods. State benefits of them. [8]

OR

Q8) a) Write steps in *Fused Deposition Modeling (FDM)* with neat sketch. State its applications. [8]

b) Write steps in *Selective Laser Sintering method* with neat sketch. State its advantages. [8]

Q9) a) Classify robot on the basis of configuration and explain Articulated configuration of Robot with advantages limitation and applications. [10]

b) Explain various joints used in Robot. [6]

OR

Q10) a) Explain the FMS system with its relevance in present era of Flexible Automation. [8]

b) Explain Group Technology. Explain the optiz coding system. [8]

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